

추론 확률 그래프의 경로 탐색을 통한 학습
(Exploring Paths in Probabilistic Graphs for Model
Training)

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Abstract

Exploring Paths in Probabilistic Graphs for Model Training

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For sequence-level training of Large Language Models (LLMs), Reinforcement Learning (RL) methods such as GRPO are widely used. However, these methods often suffer from coarse credit assignment issues, penalizing all tokens in a trajectory for a single final error, thereby overlooking potentially correct sub-paths. This paper proposes an algorithm that explores and learns from diverse probabilistic paths by exclusively targeting correct states within the training process. The proposed method applies an explicit token-level loss formulated as the expected Jensen-Shannon Divergence (JSD) or Kullback-Leibler (KL) Divergence between two distributions: one representing the selected correct token's probability and the other derived from the maximum probability token at that step. This formulation effectively encourages the model to elevate the probability of correct, non-maximal paths to match the confidence of the current greedy path. Results on the GSM8K dataset demonstrate a consistent increase in both accuracy and entropy, suggesting that the model maintains diverse reasoning capabilities while improving correctness. Furthermore, this study analyzes the training dynamics, identifying distinct phases of exploration, consolidation, and collapse, and proposes regularization terms and novel target distributions as solutions to stability limitations.

keywords : reinforcement learning, loss function design, binomial distribution, model training

1. Introduction and Motivation

The training of Large Language Models (LLMs) for complex sequence learning tasks, particularly mathematical reasoning, heavily relies on Reinforcement Learning (RL) frameworks [1]. While methods such as PPO (Proximal Policy Optimization) [10] and GRPO (Group Relative Policy Optimization) [11] have achieved significant success, they also possess inherent limitations.

1.1 Problems with Current RL Methods

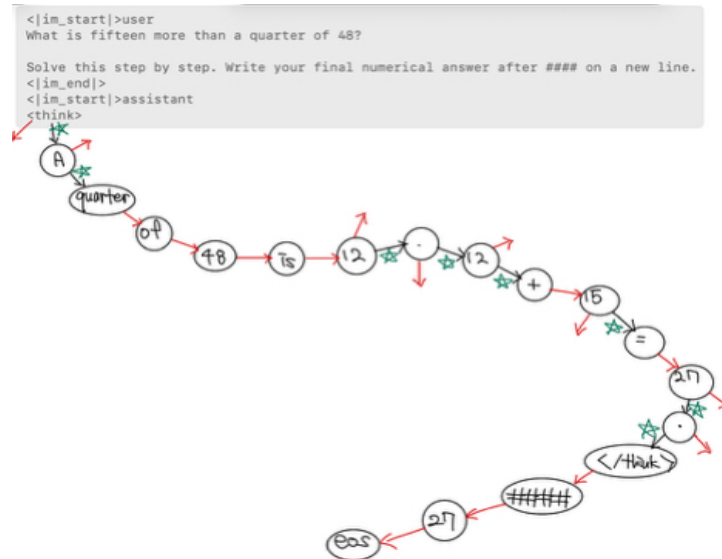
A primary issue with current outcome-based RL methods is the credit assignment problem. In tasks like math problem solving, feedback is typically sparse and binary (correct/incorrect final answer). When a generated sequence is incorrect, standard algorithms tend to penalize the entire computational path [1][2]. This approach is suboptimal because:

1. False Negatives: It overlooks the possibility that significant portions of the sub-paths (early reasoning steps) may be correct.
2. Lack of Granularity: Without a dense reward model or human expert intervention, it is difficult to pinpoint the exact wrong token.

This limitation motivates the proposed algorithm: instead of penalizing entire incorrect paths, the method selectively identifies and reinforces correct sequences to the specific target.

2. Proposed Method

This paper proposes an algorithm designed to guide the model to explore low-probability states and explicitly learn from correct sequences. The core hypothesis is that the model's inherent sampling process acts as an exploration mechanism [5]. If a sampled path, which may differ from the greedy (max-probability) path, leads to a correct answer, it constitutes a valid reasoning alternative that should be reinforced [3][4].



(Figure 1: Conceptual diagram of the learning method. Nodes represent token states. The Red Arrow indicates the max-probability path, while the Black Arrow represents the non-max probability sampled path. Black arrows leading to a correct solution are marked with green stars. Entire correct paths are Learning Targets.)

2.1 Mathematical Formulation

Let $\pi_\theta(x_t|x_{<t})$ be the probability distribution of the model at time step t , given the context $x_{<t}$. During the sampling process, let x_t^{chosen} be the token selected by the model, and x_t^{max} be the token with the highest probability in the current distribution.

If the trajectory containing x_t^{chosen} results in a correct answer, I define the loss function L which is based on the KL Divergence between two Bernoulli (Binomial) distributions.

1. Target Distribution (P): A Bernoulli distribution defined by the probability of the maximum token.

$$p_{max} = \pi_\theta(x_t^{max}|x_{<t})$$

2. Model Distribution (Q): A Bernoulli distribution defined by the probability of the chosen correct token,

$$q_{chosen} = \pi_\theta(x_t^{chosen}|x_{<t})$$

The objective is to minimize the divergence between these two distributions. The loss function L for a single step as:

$$\mathcal{L}(\theta) = \mathbb{E}_{\tau \sim \pi_{old}} \left[\mathbf{1}_{correct(\tau)} \cdot \sum_{t \in \tau} D_{KL}(\mathcal{B}(p_{max}) \parallel \mathcal{B}(q_{chosen})) \right]$$

Where $D_{KL}(\mathcal{B}(p) \parallel \mathcal{B}(q))$ is the KL divergence between two Bernoulli distributions:

$$D_{KL}(\mathcal{B}(p) \parallel \mathcal{B}(q)) = p \ln \frac{p}{q} + (1-p) \ln \frac{1-p}{1-q}$$

Properties of this Loss Design:

- Targeting the "Best Known" Confidence: The model is encouraged to raise the probability of the chosen correct token (q_{chosen}) towards the probability of the dominant token (p_{max}).
- Vanishing Loss Condition: If the chosen token is already the maximum probability token ($x_t^{chosen} = x_t^{max}$), the loss becomes 0. This prevents over-optimizing paths that are already dominant.
- Exploration Incentive: Unlike standard cross-entropy which pushes probability to 1.0, this "soft target" allows the model to maintain uncertainty and explore diverse valid paths.

3. Results and Analysis

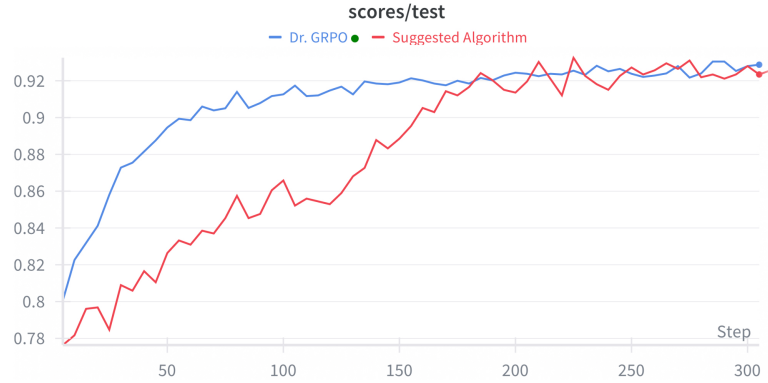
3.1 Loss Calculation using the Entire Batch

To validate this approach, experiments were conducted on the GSM8K dataset [7], a standard benchmark for mathematical reasoning.

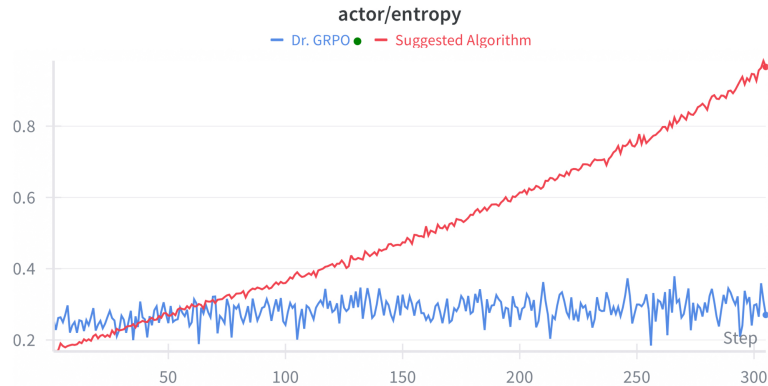
- Base Model: Qwen3-4B-Thinking-2507 [8].
- Sampling: Temperature $T=1.0$, Batch Size=512.
- Baseline: "Dr.GRPO" [12], a variant of GRPO. Configured with 4 rollouts per prompt and 128 prompts per batch to match the computational budget (512 sequences per batch).
- Parameters: Max new tokens=8192, Temperature=1.0 for training. Temperature=0 (Greedy) for evaluation.

3.1.1 Accuracy and Entropy

The proposed method demonstrates a consistent improvement in test accuracy compared to the baseline (Figure 2). Notably, the average token-level entropy also increases during training (Figure 3). The results suggest the model is effectively learning to explore a wider range of valid reasoning paths without sacrificing correctness.



(Figure 2: Test accuracy on the GSM8K dataset over training steps.)

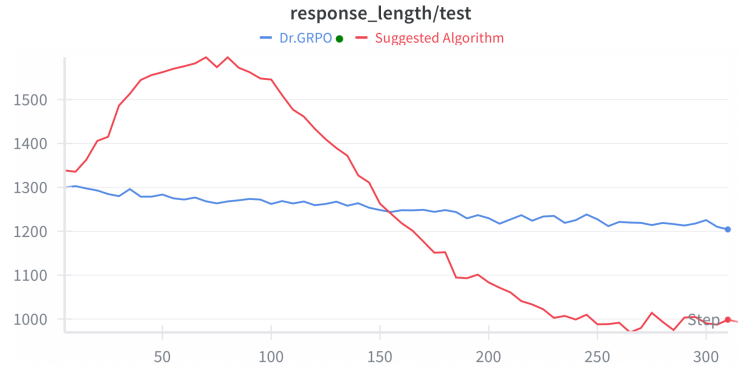


(Figure 3: Average token-level entropy during training. Entropy of states in incorrect computational path is also included.)

3.1.2 Response Length Analysis

An analysis of the average response length reveals distinct training dynamics (Figure 4):

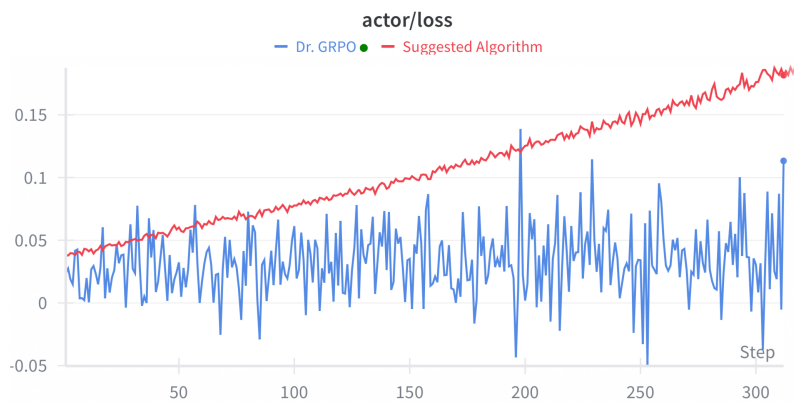
1. Phase 1: Exploration. Response length increases. The model actively explores new, longer, and more detailed computational paths to solve problems.
2. Phase 2: Consolidation. Response length decreases while accuracy rises. The model optimizes its reasoning, finding more efficient (shorter) paths among the valid ones discovered.



(Figure 4: Average response length over training steps, clearly showing the two phases: exploration and consolidation.)

3.1.3 Loss Dynamics

An unusual phenomenon was observed where the training loss increased over time (Figure 5), despite improving accuracy. This is an artifact of the loss design: the loss is calculated only for correct paths. As the model's accuracy improves, the number of "correct paths" (learning targets) in a batch increases, causing the sum of loss terms to grow. This suggests that excluding incorrect paths entirely may be more appropriate than averaging zero-loss terms.



(Figure 5: Total training loss over time. The increase reflects the growing number of correct trajectories being reinforced.)

3.1.4 The Training-Inference Discrepancy

A divergence was observed between the training and testing phases. While the model achieved high accuracy under deterministic testing conditions ($T=0$), the training accuracy under stochastic sampling ($T=1.0$) remained comparatively lower. This discrepancy suggests that while the loss function maximizes the probability of correct paths, the distribution remains too sharp for high-temperature sampling (Appendix A). Additionally, due to explorative actions, entropy increased significantly, and the training score began to decrease after approximately 250 steps.

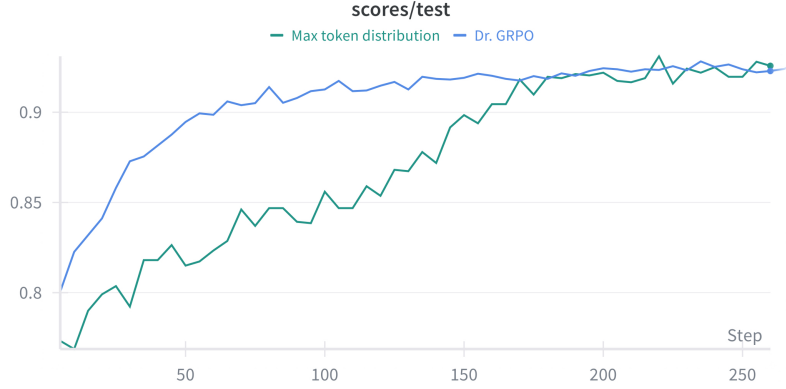
3.2 Using Only correct sequences for loss

To mitigate the increasing loss issue, the loss function was normalized by the number of correct sequences rather than the total batch size. Except for the normalization, all settings remained identical.

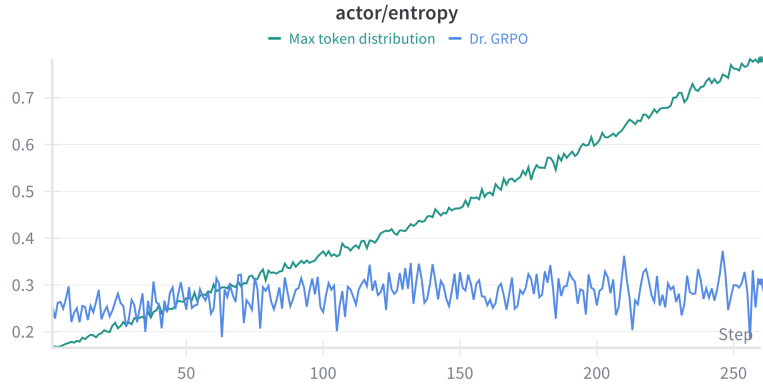
$$\begin{aligned}\mathcal{L}(\theta) &= \frac{\mathbb{E}_{\tau \sim \pi_{\text{old}}} \left[\mathbf{1}_{\text{correct}(\tau)} \cdot \sum_{t \in \tau} D_{\text{KL}}(\mathcal{B}(p_{\text{max}}) \parallel \mathcal{B}(q_{\text{chosen}})) \right]}{\mathbb{E}_{\tau \sim \pi_{\text{old}}} \left[\mathbf{1}_{\text{correct}(\tau)} \right]} \\ &= \mathbb{E}_{\tau \sim \pi_{\text{old}}(\cdot | \text{correct})} \left[\sum_{t \in \tau} D_{\text{KL}}(\mathcal{B}(p_{\text{max}}) \parallel \mathcal{B}(q_{\text{chosen}})) \right]\end{aligned}$$

3.2.1 Accuracy and Entropy

The score and entropy curves were similar to Section 3.1.



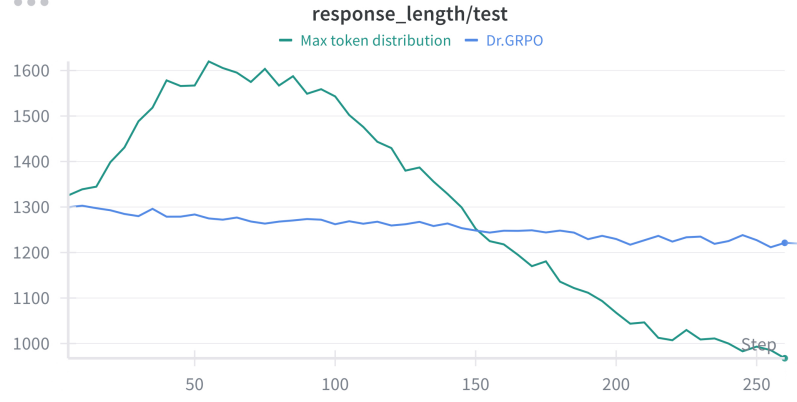
(Figure 6: Test accuracy on the GSM8K dataset over training steps.)



(Figure 7: Average token-level entropy during training. Only entropy of states in correct computational paths is included.)

3.2.2 Response Length Analysis

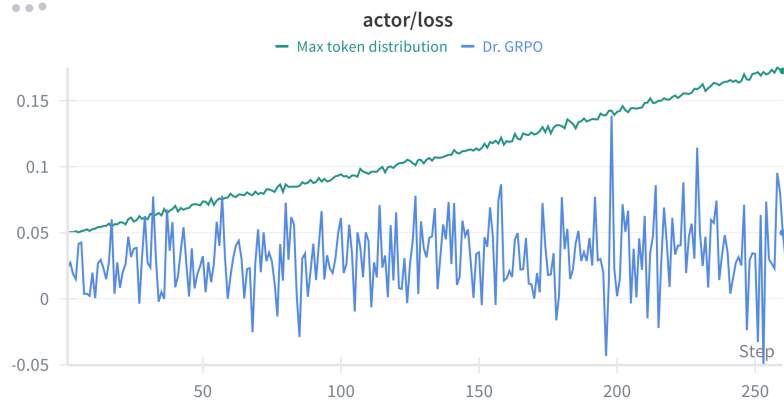
Similar to 3.1, the model trained with the max-probability token distribution used significantly fewer tokens than the baseline Dr.GRPO model.



(Figure 8: Average response length over training steps.)

3.2.3 Loss Dynamics

Normalizing the loss function did not help reduce the aggregate loss value. It appears that as the model explores more, the divergence values themselves increase.



(Figure 9: Total training loss over time.)

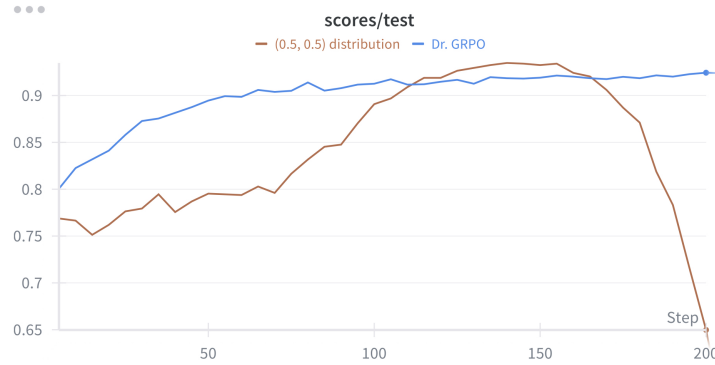
3.3 Using a Fixed Soft Target (0.5, 0.5)

In this experiment, I used different target distribution. Instead of forcing the correct token's probability to chase the maximum probability for each states, a soft fixed target was employed. By anchoring the correct token's probability to a fixed value of 0.5, the model is more strongly induced to select correct paths.

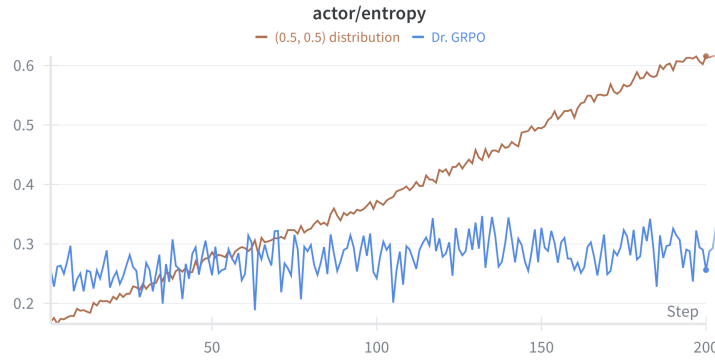
$$\begin{aligned}
 \mathcal{L}(\theta) &= \frac{\mathbb{E}_{\tau \sim \pi_{\text{old}}} \left[\mathbf{1}_{\text{correct}(\tau)} \cdot \sum_{t \in \tau} D_{\text{KL}}(\mathcal{B}(0.5) \parallel \mathcal{B}(q_{\text{chosen}})) \right]}{\mathbb{E}_{\tau \sim \pi_{\text{old}}} \left[\mathbf{1}_{\text{correct}(\tau)} \right]} \\
 &= \mathbb{E}_{\tau \sim \pi_{\text{old}}(\cdot | \text{correct})} \left[\sum_{t \in \tau} D_{\text{KL}}(\mathcal{B}(0.5) \parallel \mathcal{B}(q_{\text{chosen}})) \right]
 \end{aligned}$$

3.3.1 Accuracy and Entropy

Test accuracy improved more rapidly than in the method which is explained in Section 3.1, as the incentive for selecting the correct path was stronger. However, after reaching a peak, the model's performance collapsed. Although the training loss consistently decreased until 200 steps, the subsequent collapse suggests an instability in the loss formulation or an overfitting to the fixed target.



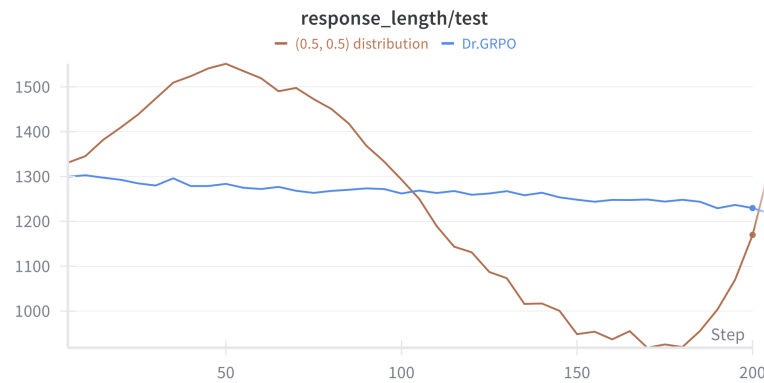
(Figure 10: Test accuracy on the GSM8K dataset over training steps.)



(Figure 11: Average token-level entropy during training. Only entropy of states in correct computational paths is included.)

3.3.2 Response Length Analysis

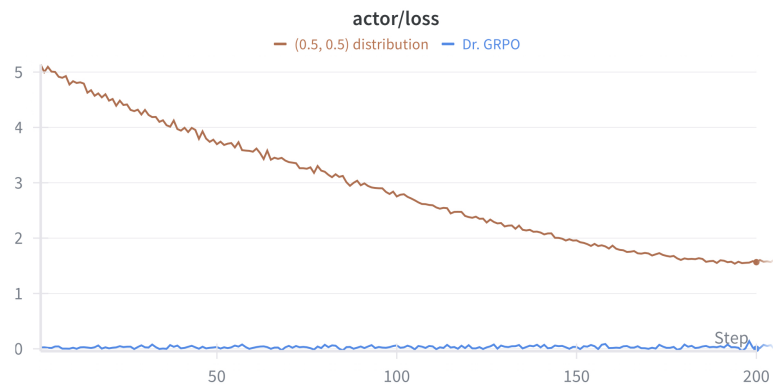
Before the collapse, the average response length curve followed a trend similar to Section 3.1. Notably, at the step of maximum test accuracy, the model utilized significantly fewer tokens, mirroring the efficiency gains observed in Section 3.1.



(Figure 12: Average response length over training steps.)

3.3.3 Loss Dynamics

Unlike Section 3.1, the training loss consistently decreased throughout the training process. However, the fact that the test score collapsed while the loss continued to decrease indicates that the loss function may not be perfectly aligned with the evaluation metric or may be prone to reward hacking, leading to distribution collapse.



(Figure 13: Total training loss over time.)

4. Limitations and Future Work

4.1 Limitations

Initially, test accuracy improved significantly but eventually collapsed. This suggests that the current loss function may not be sustainable for continuous, long-term learning.

4.2 Proposed Solution 1: Entropy Regularization

To address the stability issue, I propose introducing an Entropy Regularization term. I am currently conducting experiments adding a penalty to the loss function to dampen uncontrolled entropy growth:

$$\mathcal{L}_{total} = \mathcal{L}_{KL} + \lambda \cdot \mathcal{H}(\pi_{\theta})$$

Future trials will investigate the sensitivity of λ at values.

4.3 Proposed Solution 2: Rank-Aware Target Distribution with JSD

To further refine the learning signal, I propose a new loss function based on Jensen-Shannon Divergence (JSD) with a Rank-Aware Target Distribution. Instead of a simple binary target, I construct a target distribution p_{target} derived from the current model distribution $q_{current}$.

If the selected correct token x_{chosen} has a rank k in the current distribution (sorted by probability), the target distribution is constructed by promoting x_{chosen} to rank 1, and shifting the original top $k-1$ tokens down by one rank.

$$p_{target}(x) = \begin{cases} q_{(1)} & \text{if } x = x_{chosen} \\ q_{(r+1)} & \text{if } \text{rank}(x) = r < k \\ q(x) & \text{if } \text{rank}(x) > k \end{cases}$$

The loss is then defined as the JSD between this target and the current distribution:

$$\mathcal{L}(\theta) = \mathbb{E}_{\tau \sim \pi_{old}} \left[\mathbf{1}_{correct(\tau)} \cdot \sum_{t \in \tau} D_{JSD}(p_{target} \parallel q_{current}) \right]$$

Computational Efficiency: A key advantage of this formulation is computational efficiency. Since $p_{target}(x) = q_{current}(x)$ for all tokens with rank $> k$, the contribution to the JSD from these tails is exactly zero.

$$p_i = q_i \implies p_i \log \frac{p_i}{(p_i + q_i)/2} = 0$$

Therefore, the loss calculation only requires processing the top k tokens (where k is the rank of the chosen token, typically small, e.g., $k \leq 20$), reducing the complexity from $O(|V|)$ to $O(k)$. This allows for efficient training even with large vocabularies.

4.4 Future work: Pre-training Application

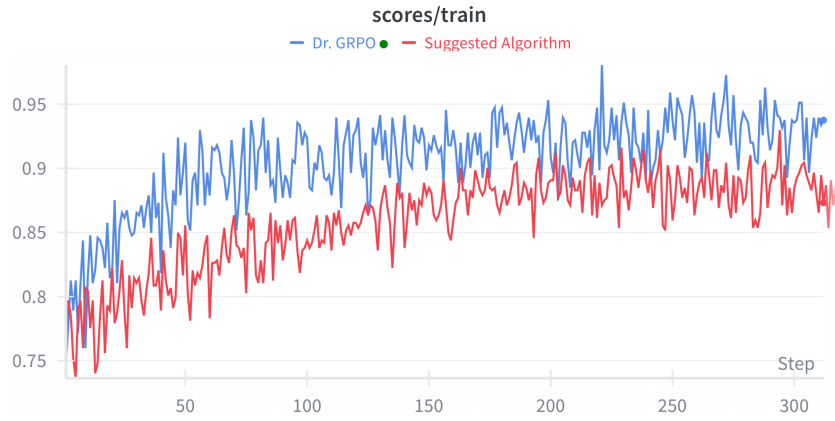
I anticipate applying this greedy JSD loss to the pre-training phase to encourage the model to continuously attempt to assign the highest probability to correct tokens.

References

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Appendix

Appendix A



(Figure A1: The training loss trajectory. The existence of a gap of trend in training score, despite very similar test score, highlights the distributional gap between the model's stochastic exploration (training) and its deterministic exploitation (testing).

Code

Code is available at github.com/jsbyun121/exploring-paths-llm

국문초록

현재 거대 언어 모델의 시퀀스 단위 학습에서 주로 사용되는 GRPO 등의 강화학습 기법은 종종 잘못된 추론 경로에 대해 해당 경로 상의 모든 토큰을 처벌함으로써, 하위 경로가 올바른 경우를 간과하는 신용 할당(Credit Assignment) 문제를 겪는다. 본 논문에서는 올바른 상태만을 학습 대상으로 삼아 다양한 확률적 경로를 탐색하고 학습하는 새로운 알고리즘을 제안한다. 제안하는 방법은 기존의 확률적 경로에 대해 명시적인 토큰 수준 손실을 적용한다. 이 손실 함수는 선택된 정답 토큰의 이항 분포와 해당 시점의 최대 확률 토큰의 이항 분포 간의 기대 KL 발산(또는 JSD)으로 공식화되어, 모델이 올바른 경로의 확률을 높이도록 효과적으로 유도한다. 이 접근법은 올바른 계산 경로가 더 높은 확률을 갖도록 자연스럽게 유도하는 동시에, 모델 추론 시 진행되는 샘플링 과정 자체가 새로운 경로로의 탐색을 촉진한다. GSM8K 데이터셋에서의 실험 결과는 정확도와 엔트로피 모두에서 일관된 증가를 보여주었으며, 이는 모델이 정확성을 개선하면서도 다양한 사고 능력을 유지함을 시사한다. 또한 본 연구에서는 학습 과정을 탐색(Exploration), 강화(Consolidation), 붕괴(Collapse)의 단계로 분석하고, 안정성 문제를 해결하기 위한 새로운 순위 기반 목표 분포(Rank-Aware Target Distribution)와 정규화 방안을 제시한다.

주요어 : 강화학습, 손실 함수 설계, 이항 분포, 모델 학습